BRCA-TCGA — foundation-model performance

Main classifiers — Accuracy

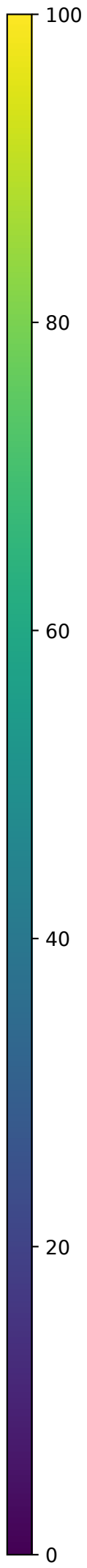
	Omiclass	Logistic Regression
DE baseline		
DE	86.4 ±1.2	N/A
DE · corrected	88.5 ±1.7	N/A
BulkFormer		
Full	84.9	84.3
Full · corrected	76.4	80.6
PCA	84.3	86.2
PCA · corrected	74.3	81.9
Supervised	83.6	81.1
Supervised · corrected	73.6	81.2
Supervised + PCA	79.9	82.4
Supervised + PCA · corrected	72.9	79.9
CellFM 1-view		
Full	53.5	59.7
Full · corrected	48.6	51.4
PCA	55.3	48.4
PCA · corrected	59.0	51.4
Supervised	50.3	42.1
Supervised · corrected	47.2	41.0
Supervised + PCA	49.7	48.4
Supervised + PCA · corrected	52.8	49.3
CellFM 20-view		
Full	81.8	82.4
Full · corrected	80.6	84.0
PCA	80.5	80.5
PCA · corrected	84.7	81.9
Supervised	82.4	78.0
Supervised · corrected	78.5	79.9
Supervised + PCA	79.2	74.8
Supervised + PCA · corrected	79.2	74.3
BulkRNABert		
Full	76.7	79.2
Full · corrected	75.7	77.8
PCA	79.2	80.5
PCA · corrected	70.8	76.4
Supervised	77.4	79.9
Supervised · corrected	75.0	76.4
Supervised + PCA	80.5	79.2
Supervised + PCA · corrected	75.7	73.6

Main classifiers — Macro-F1

	Omiclass	Logistic Regression
DE baseline		
DE	83.3 ±1.1	N/A
DE · corrected	85.4 ±2.2	N/A
BulkFormer		
Full	82.0	81.2
Full · corrected	72.8	77.8
PCA	80.8	83.6
PCA · corrected	70.4	78.9
Supervised	75.3	74.9
Supervised · corrected	68.1	78.8
Supervised + PCA	74.7	78.1
Supervised + PCA · corrected	64.1	75.4
CellFM 1-view		
Full	41.5	47.2
Full · corrected	28.0	37.3
PCA	34.4	41.6
PCA · corrected	35.2	44.4
Supervised	20.7	33.2
Supervised · corrected	34.8	35.4
Supervised + PCA	25.5	38.0
Supervised + PCA · corrected	22.3	41.7
CellFM 20-view		
Full	79.5	79.5
Full · corrected	76.0	79.9
PCA	77.9	78.0
PCA · corrected	80.8	78.9
Supervised	79.1	76.5
Supervised · corrected	72.5	75.5
Supervised + PCA	75.9	70.4
Supervised + PCA · corrected	73.0	73.7
BulkRNABert		
Full	66.8	69.5
Full · corrected	70.0	72.4
PCA	68.9	74.1
PCA · corrected	62.1	70.3
Supervised	68.5	70.1
Supervised · corrected	67.5	70.9
Supervised + PCA	70.9	73.8
Supervised + PCA · corrected	71.4	69.0

Additional classifiers — Accuracy

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	83.6	84.9	78.6
Full · corrected	81.2	84.7	79.2
PCA	85.5	83.0	83.0
PCA · corrected	81.2	82.6	79.9
Supervised	83.6	81.1	80.5
Supervised · corrected	80.6	81.2	80.6
Supervised + PCA	82.4	81.1	79.9
Supervised + PCA · corrected	76.4	76.4	79.9
Full tokens	84.3	N/A	N/A
Full tokens · corrected	79.2	N/A	N/A
CellFM 1-view			
Full	56.6	54.7	50.9
Full · corrected	52.8	56.9	50.7
PCA	56.0	52.8	58.5
PCA · corrected	54.9	57.6	57.6
Supervised	49.1	45.9	54.1
Supervised · corrected	43.8	41.7	47.2
Supervised + PCA	49.1	49.7	56.6
Supervised + PCA · corrected	41.0	41.0	45.1
Full tokens	45.3	N/A	N/A
Full tokens · corrected	41.7	N/A	N/A
CellFM 20-view			
Full	74.8	76.1	78.0
Full · corrected	77.1	77.8	77.8
PCA	83.0	83.6	81.8
PCA · corrected	81.2	81.9	77.8
Supervised	76.7	75.5	79.2
Supervised · corrected	77.1	69.4	81.2
Supervised + PCA	76.1	77.4	78.0
Supervised + PCA · corrected	79.2	76.4	76.4
Full tokens	67.9	N/A	N/A
Full tokens · corrected	70.8	N/A	N/A
BulkRNABert			
Full	78.6	76.1	78.0
Full · corrected	77.1	74.3	75.0
PCA	81.8	83.0	81.8
PCA · corrected	79.9	76.4	75.7
Supervised	79.9	76.7	78.0
Supervised · corrected	76.4	74.3	74.3
Supervised + PCA	81.8	78.0	79.9
Supervised + PCA · corrected	78.5	78.5	75.0

Additional classifiers — Macro-F1

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	80.1	79.6	73.5
Full · corrected	80.1	83.7	76.8
PCA	83.9	80.2	77.9
PCA · corrected	78.4	81.3	76.2
Supervised	81.2	76.5	77.2
Supervised · corrected	77.6	78.1	76.5
Supervised + PCA	78.1	74.9	75.1
Supervised + PCA · corrected	72.2	72.5	74.7
Full tokens	83.5	N/A	N/A
Full tokens · corrected	78.4	N/A	N/A
CellFM 1-view			
Full	41.5	42.2	16.9
Full · corrected	36.6	41.7	16.8
PCA	45.8	46.2	33.0
PCA · corrected	44.4	46.5	34.6
Supervised	35.8	32.7	29.4
Supervised · corrected	33.9	31.3	22.6
Supervised + PCA	39.8	38.8	39.5
Supervised + PCA · corrected	31.5	30.8	27.6
Full tokens	24.2	N/A	N/A
Full tokens · corrected	20.6	N/A	N/A
CellFM 20-view			
Full	55.4	71.4	65.7
Full · corrected	68.4	69.1	65.5
PCA	80.2	80.8	78.9
PCA · corrected	78.9	78.9	72.5
Supervised	71.4	67.3	73.3
Supervised · corrected	68.5	63.8	73.9
Supervised + PCA	73.4	75.3	72.4
Supervised + PCA · corrected	75.3	73.6	71.0
Full tokens	44.8	N/A	N/A
Full tokens · corrected	50.1	N/A	N/A
BulkRNABert			
Full	70.8	69.7	69.5
Full · corrected	70.2	67.1	67.3
PCA	74.9	76.5	76.2
PCA · corrected	72.7	72.3	63.7
Supervised	68.3	70.3	68.6
Supervised · corrected	69.9	67.9	65.1
Supervised + PCA	74.5	71.3	70.3
Supervised + PCA · corrected	71.3	73.9	62.6